DGMS 2019 • SECTION J: ORE GEOLOGY & ECONOMIC GEOLOGY

Question 76: Supergene Copper Deposit Weathering & Geochemical Zoning

Official Examination Guide • Supergene Sulfide Enrichment vs. Oxidized Leached Profiles

OFFICIAL EXAMINATION QUESTION

Which one of the following statements regarding supergene copper deposit is NOT correct?

(a) The top zone consisting of goethitic/limonitic mass with some clay is gossan.

(b) The zone of secondary enrichment lies below the groundwater surface where copper content increases.

(c) The supergene enrichment zone gradually merges with the protore.

(d) The secondary enrichment zone contains native copper and chalcocite. CORRECT ANSWER (INCORRECT STATEMENT)

1. Executive Summary & Core Principle

When a primary hypogene copper sulfide ore body (such as a porphyry copper deposit containing low-grade **chalcopyrite (CuFeS_2)** and **pyrite (FeS_2)**) is exposed to subaerial weathering and meteoric water

infiltration, it develops a pronounced vertical geochemical and mineralogical zoning profile:

- **Gossan (Leached Capping):** The uppermost, highly porous, sponge-like residue composed of goethite, limonite, hematite, jarosite, residual quartz, and clays. Soluble copper has been completely leached and transported downward.
- **Oxidized Zone (Vadose Zone above Water Table):** Oxygenated zone where acidic, metal-bearing solutions react with carbonates and host rock silicates, precipitating copper carbonates (malachite, azurite), silicates (chrysocolla), oxides (cuprite), and **native copper (Cu^0)**. Native copper forms under reducing microsites in the oxidized zone or at the fluctuating vadose-phreatic boundary where copper sulfate solutions are reduced by ferrous sulfate (FeSO_4) or organic matter.
- **Supergene Sulfide Enrichment Zone (Secondary Enrichment Blanket):** Situated at and directly below the water table (phreatic surface). In this reducing, oxygen-depleted, and pH-neutralized environment, descending copper-rich sulfate solutions react with primary hypogene sulfides via cation exchange/displacement, precipitating secondary **copper sulfides**—principally **chalcocite (Cu_2S)**, **covellite (CuS)**, and **digenite (Cu_9S_5)**. This zone dramatically upgrades ore grades from 0.3–0.5% in the primary protore to 2–5% Cu.
- **Protore (Primary Hypogene Zone):** The unaltered, deep primary hydrothermal mineral deposit. The supergene sulfide enrichment zone grades irregularly and **gradually merges with the protore** as descending copper solutions become exhausted with depth.

Therefore, statement **(d)** is **NOT correct**: native copper is an oxidation-zone / vadose mineral precipitate, whereas the secondary enrichment zone is characterized specifically by **secondary copper sulfide minerals** (chalcocite, covellite), not native copper.

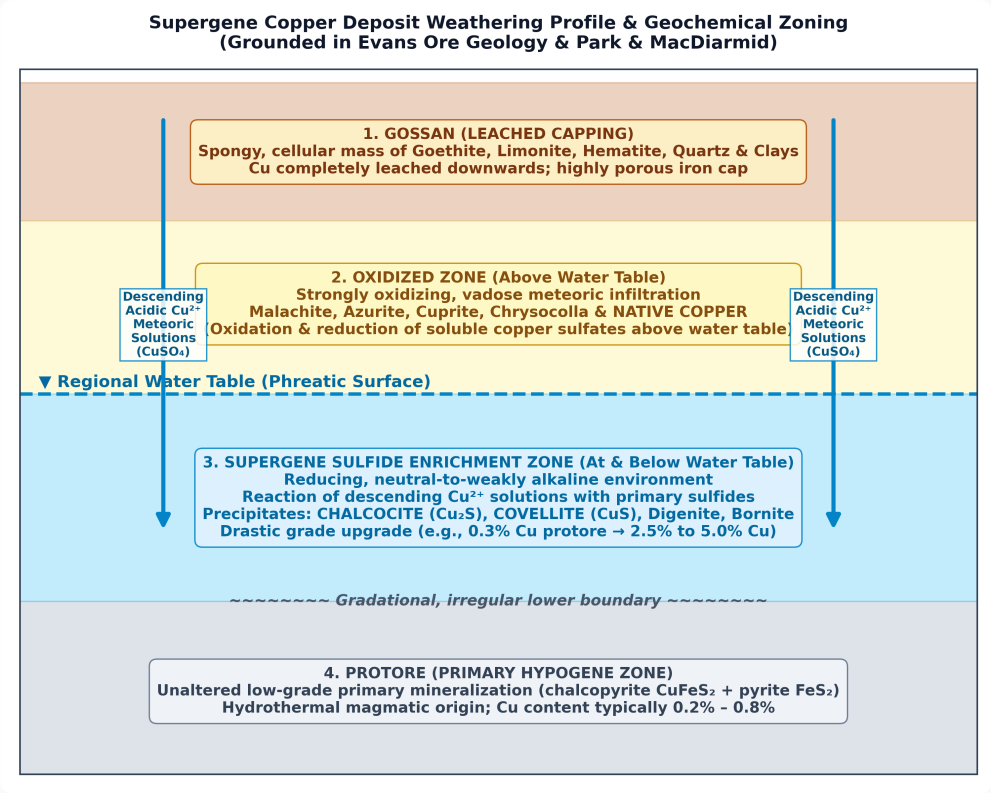


*Figure 76.1: Vertical profile and geochemical zoning of a supergene copper deposit, illustrating the gossan, oxidized vadose zone, water table, supergene sulfide enrichment blanket, and gradational merger into primary hypogene protore (Grounded in Evans, *Ore Geology and Industrial Minerals* and Park & MacDiarmid).*

2. Geochemical Reactions Across the Weathering Profile

The formation of a supergene copper deposit involves distinct geochemical stages:

A. Oxidation and Leaching (Gossan & Oxidized Zone)

Pyrite (FeS_2) oxidation by meteoric water and atmospheric oxygen generates sulfuric acid (H_2SO_4) and ferric sulfate ($\text{Fe}_2(\text{SO}_4)_3$), the two most potent natural leaching agents:

$$2FeS_2 + 7O_2 + 2H_2O \rightarrow 2FeSO_4 + 2H_2SO_4$$
$$4FeSO_4 + O_2 + 2H_2SO_4 \rightarrow 2Fe_2(SO_4)_3 + 2H_2O$$

Ferric sulfate vigorously leaches primary chalcopyrite, putting copper into solution as soluble copper sulfate (CuSO_4):

$$CuFeS_2 + 2Fe_2(SO_4)_3 \rightarrow CuSO_4 + 5FeSO_4 + 2S^0$$

Iron hydrolyzes and precipitates in the upper vadose zone as insoluble goethite and limonite, forming the **gossan**:

$$Fe_2(SO_4)_3 + 6H_2O \rightarrow 2Fe(OH)_3\downarrow + 3H_2SO_4$$
$$Fe(OH)_3 \rightarrow FeO(OH)\ (Goethite) + H_2O$$

B. Supergene Sulfide Enrichment (Below Water Table)

As the acidic, copper-rich solution percolates below the phreatic surface (water table), atmospheric oxygen is excluded. The copper sulfate solutions react with primary hypogene sulfides. Because copper sulfide is far less soluble than iron sulfide (Schürmann's series), Cu^{2+} ions replace Fe^{2+}:

$$5CuSO_4 + CuFeS_2\ (Chalcopyrite) + 4H_2O \rightarrow 3Cu_2S\ (Chalcocite) + FeSO_4 + 4H_2SO_4$$
$$7CuSO_4 + 4FeS_2\ (Pyrite) + 4H_2O \rightarrow 7CuS\ (Covellite) + 4FeSO_4 + 4H_2SO_4$$

This replacement creates soot-like or massive coatings of **chalcocite (Cu_2S)** and **covellite (CuS)** over primary grains, transforming an unworkable 0.4% Cu protore into a highly profitable deposit.

3. Comprehensive Comparative Zonal Analysis

Zone	Position Relative to Water Table	Dominant Minerals	Copper Grade & Behavior
1. Gossan (Leached Capping)	Surface to near vadose zone	Goethite ($\alpha\text{-FeO(OH)}$), Limonite,	Extremely low (<0.05% Cu); copper is completely leached and

Zone	Position Relative to Water Table	Dominant Minerals	Copper Grade & Behavior
		Hematite, Jarosite, Quartz, Clays	evacuated downward.
2. Oxidized Zone	Vadose zone (above water table)	Malachite, Azurite, Cuprite (Cu_2O), Chrysocolla, *Native Copper (Cu^0)*	Moderate (0.5–1.5% Cu); trapped by carbonates, silicates, and localized reducing agents.
3. Supergene Enrichment Blanket	At and immediately below water table	**Chalcocite (Cu_2S)**, **Covellite (CuS)**, Digenite, Bornite	**Peak Grade (2.0–5.0% Cu)**; massive economic enrichment via sulfide replacement.
4. Protore (Hypogene Zone)	Deep beneath weathering penetration	Chalcopyrite (CuFeS_2), Pyrite (FeS_2), Bornite	Primary baseline grade (typically 0.2–0.8% Cu); hydrothermal origin.

4. Systematic Option Evaluation

- **Statement (a) is CORRECT:** "The top zone consisting of goethitic/limonitic mass with some clay is gossan."
  Geological Reality: By definition, a gossan (German: *Gössan*; French: *chapeau de fer*) is the oxidized iron-hydroxide cellular capping formed by the subaerial weathering of sulfide deposits.
- **Statement (b) is CORRECT:** "The zone of secondary enrichment lies below the groundwater surface where copper content increases."
  Geological Reality: The exclusion of oxygen below the water table provides the reducing conditions essential for precipitating secondary copper sulfides, concentrating copper into an enrichment blanket.

- **Statement (c) is CORRECT:** "The supergene enrichment zone gradually merges with the protore."
  Geological Reality: With increasing depth below the water table, descending meteoric solutions become spent and rock permeability decreases along fractures; thus, the degree of replacement progressively drops until the enriched zone seamlessly and gradually transitions into unaltered protore.
- **Statement (d) is NOT CORRECT:** "The secondary enrichment zone contains native copper and chalcocite."
  Geological Reality: The secondary enrichment zone is defined by **secondary copper sulfides** (chalcocite, covellite, digenite). Native copper (Cu^0) belongs to the **oxidized zone** (formed by the reduction of copper sulfate by ferrous iron or near-surface organic carbon) and does not define the secondary sulfide enrichment blanket.

> **Key Takeaway for DGMS / UPSC Geologist Exams:**
> The supergene copper profile consists of:
>
> **Gossan (Leached) → Oxidized Zone (Carbonates/Oxides/Native Cu) → Water Table → Supergene Sulfide Blanket (Chalcocite/Covellite) → Gradational Merger into Protore (Chalcopyrite/Pyrite).**

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